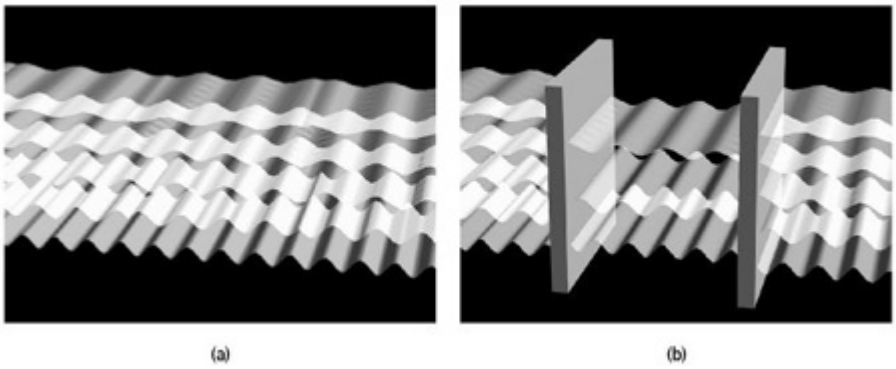


Figure 12.1 **(**a**)** Vacuum fluctuations of the electromagnetic field. **(**b**)** Vacuum fluctuations between two metal plates and those outside the plates.

Think about how thoroughly odd this is. You place two plain, ordinary, uncharged metal plates into an *empty* region of space, facing one another. As their masses are tiny, the gravitational attraction between them is so small that it can be completely ignored. Since there is nothing else around, you naturally conclude that the plates will stay put. But this is *not* what Casimir's calculations predicted would happen. He concluded that the plates would be gently guided by the ghostly grip of quantum vacuum fluctuations to move toward one another.

When Casimir first announced these theoretical results, equipment sensitive enough to test his predictions didn't exist. Yet, within about a decade, another Dutch physicist, Marcus Spaarnay, was able to initiate the first rudimentary tests of this *Casimir force*, and increasingly precise experiments have been carried out ever since. In 1997, for example, Steve Lamoreaux, then at the University of Washington, confirmed Casimir's predictions to an accuracy of 5 percent.[4] (For plates roughly the size of playing cards and placed one ten-thousandth of a centimeter apart, the force between them is about equal to the weight of a single teardrop; this shows how challenging it is to measure the Casimir force.) There is now little doubt that the intuitive notion of empty space as a static, calm, eventless arena is thoroughly off base. Because of quantum uncertainty, empty space is teeming with quantum activity.

It took scientists the better part of the twentieth century to fully develop the mathematics for describing such quantum activity of the electromagnetic, and strong and weak nuclear forces. The effort was well